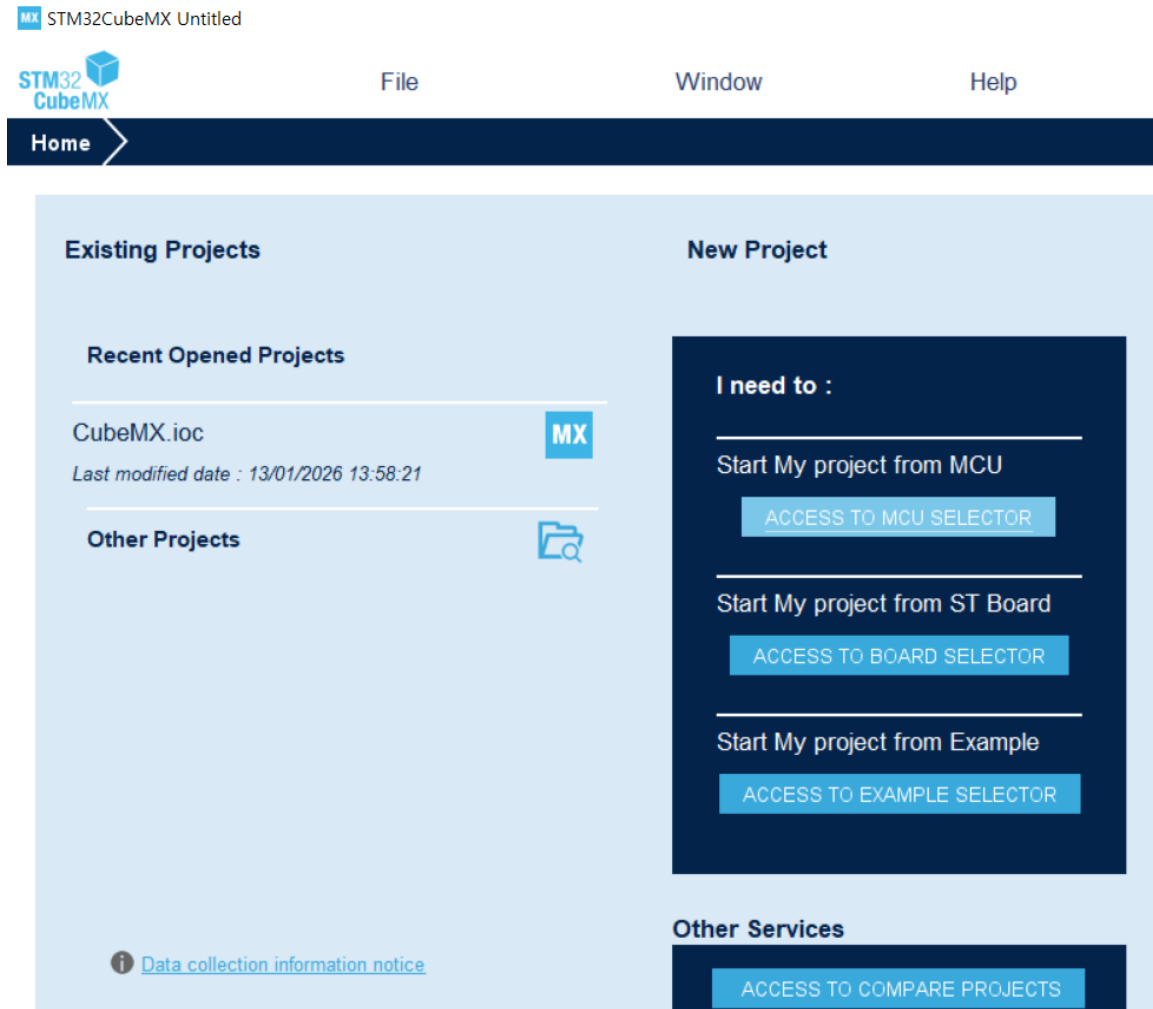


BNO055 – 9축 센서 실습

2026.06.17

Project 생성

- STM32CubeMX 실행 후 "ACCESS TO MCU SELECTOR" 선택



Project 생성

- STM32F407ZET6 선택 후 "Start Project" 클릭

MX New Project from a MCU/MPU

MCU/MPU Selector Board Selector Example Selector Cross Selector

MCU/MPU Filters

Commercial Part Number STM32F407ZET6

PRODUCT INFO

- Segment
- Series
- Line
- Marketing Status
- Price
- Package
- Core
- Coprocessor

STM32F4 Series

STM32F407ZET6

High-performance foundation line, Arm Cortex-M4 core with DSP and FPU, 512 Kbytes of Flash memory, 168 MHz CPU, ART Accelerator, Ethernet, FSMC

ACTIVE
Product is in mass production

Unit Price for 10kU (US\$): 5.605

4 mm LQFP 144 20x20x1.

The STM32F405xx and STM32F407xx family is based on the high-performance

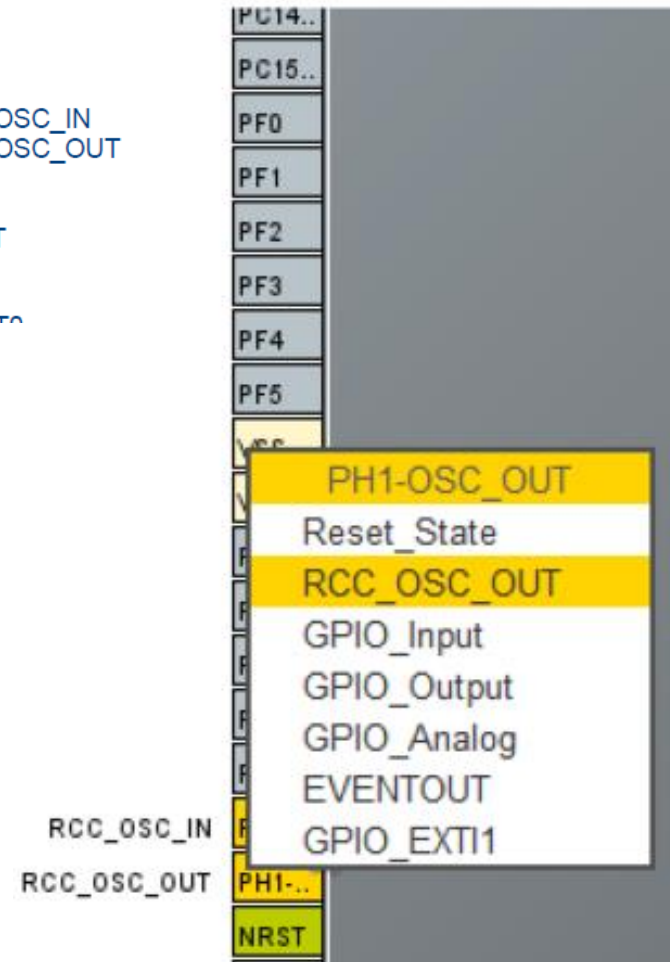
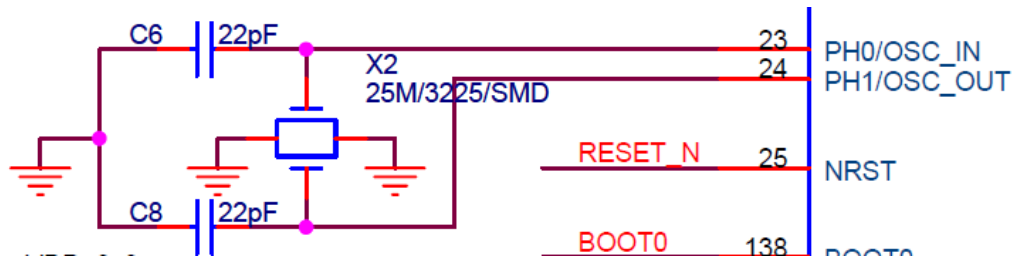
MCUs/MPUs List: 2 items

* Comm...	Part No	Reference	Mark...	Unit Pric...	B...	Package	Flash	RAM	...	Fre...
☆ STM32F4...	STM32...	STM32...	Active	5.605		LQFP 144 20x...	512 ...	192 ...	1...	168 ...
☆ STM32F...	STM32...	STM32...	Active	5.605		LQFP 144 20x...	512 ...	192 ...	1...	168 ...

Export

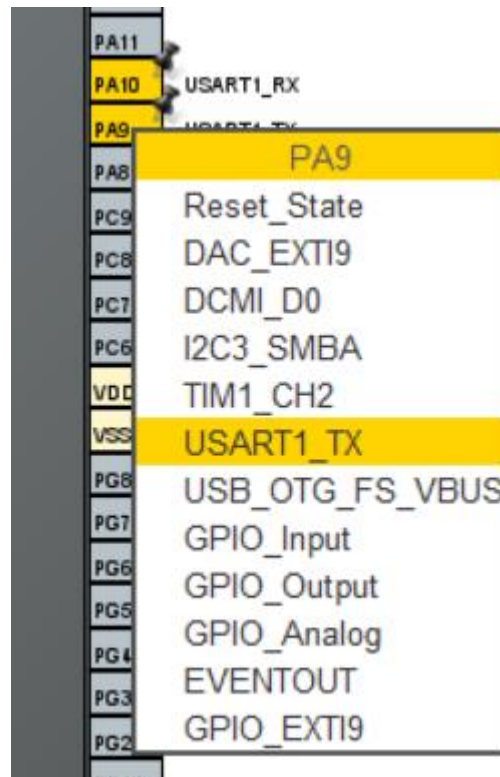
Project 생성

- HSE(25MHz) 핀 설정 - 회로도 참조
- HSE : PH0-OSC_IN, PH1-OSC_OUT



Project 생성

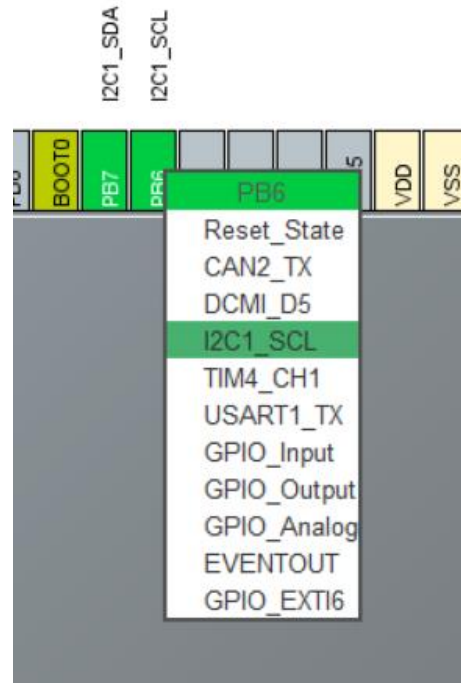
- USART1핀 설정 - 회로도 참조
- UART1 : PA9-USART1_TXD, PA10-USART1_RXD



Project 생성

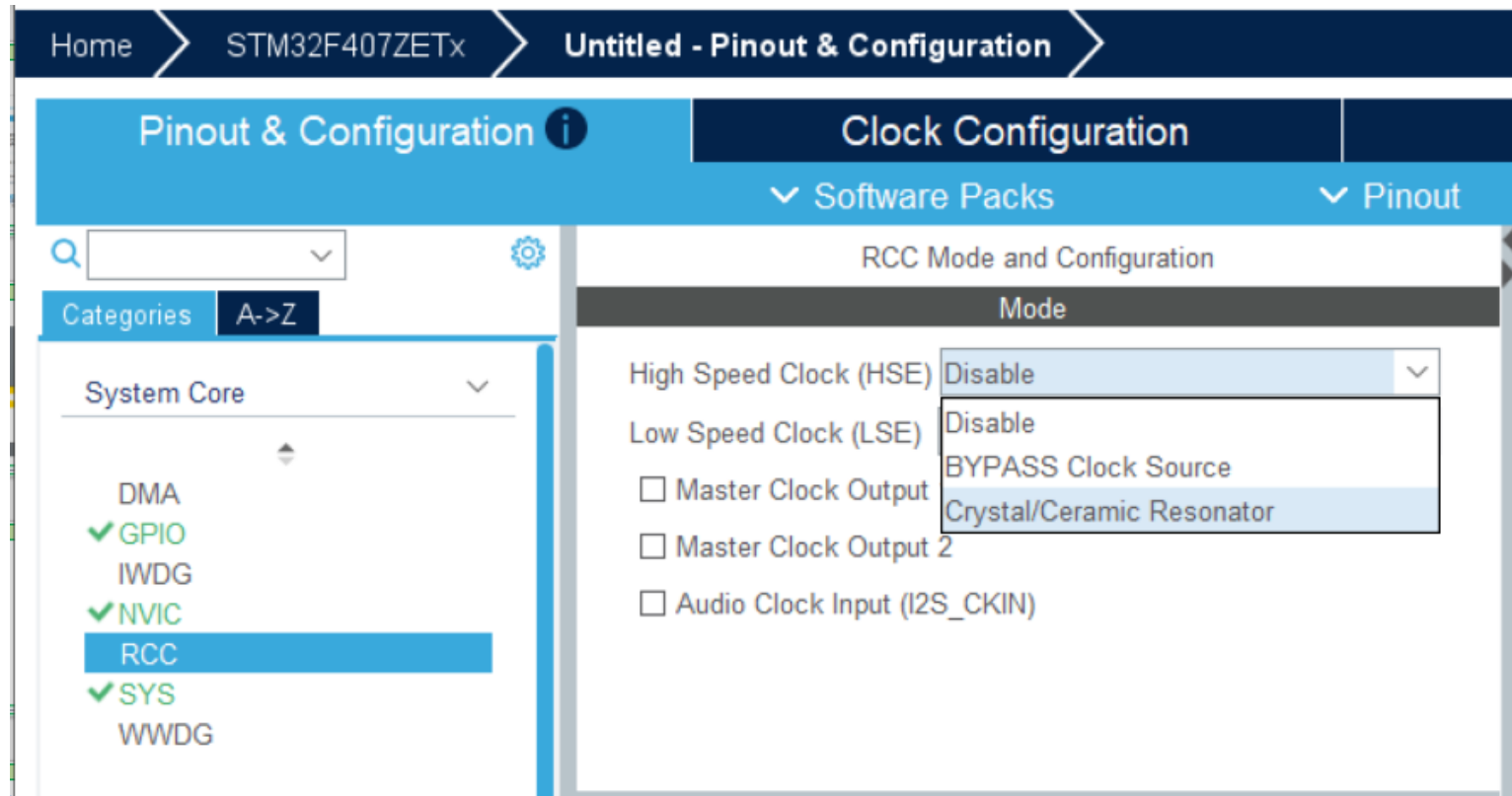
- I2C 핀 설정 - 회로도 참조
- I2C : PB6-I2C1_SCL, PB7-I2C1_SDA

PB5	136	I2C1_SCL
PB6	137	I2C1_SDA
PB7	139	CAN1_RX



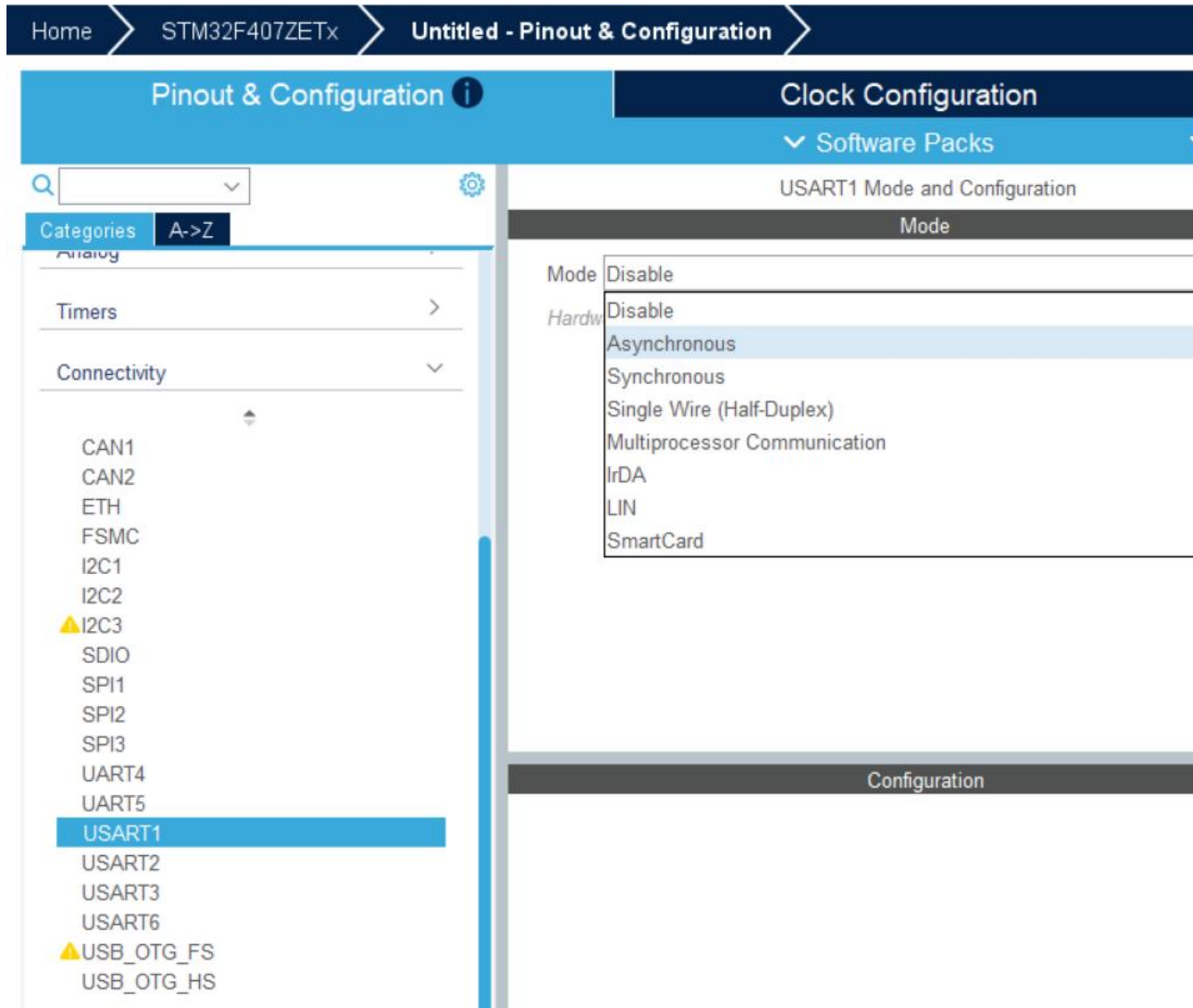
Project 생성

- System Core -> RCC에서 HSE 활성화 -> Crystal/Ceramic Resonator 선택



Project 생성

- Connectivity -> USART1 -> Asynchronous 선택



Project 생성

- Connectivity -> I2C1 -> Mode -> I2C 선택

I2C1 Mode and Configuration

Mode

I2C I2C

Configuration

Reset Configuration

✓ NVIC Settings ✓ DMA Settings ✓ GPIO Settings

✓ Parameter Settings ✓ User Constants

Configure the below parameters :

Search (Ctrl+F)

✓ Master Features

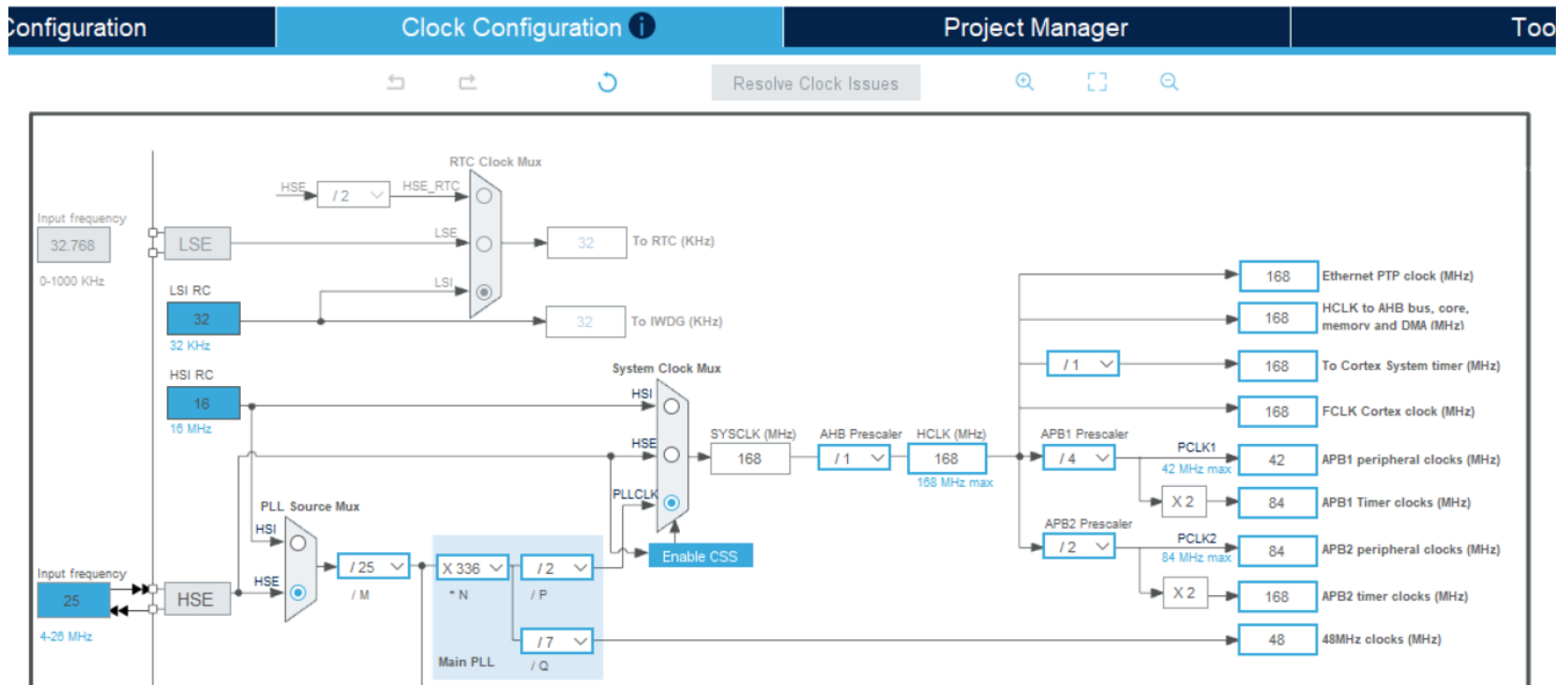
I2C Speed Mode	Standard Mode
I2C Clock Speed (Hz)	100000

✓ Slave Features

Clock No Stretch Mode	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0
General Call address detection	Disabled

Project 생성

- Clock Configuration **클릭** -> Yes **클릭**



Project 생성

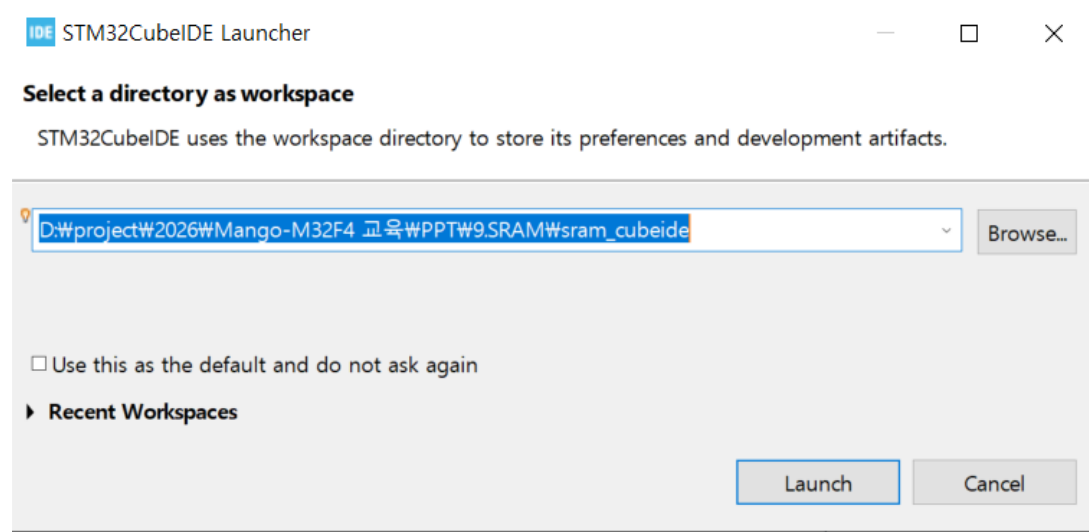
- Project Manager에서 Project Name 기입, IDE를 STM32CubeIDE로 선택 후 GENERATE CODE 클릭

The screenshot shows the STM32CubeIDE Project Manager interface. The top navigation bar includes 'Home', 'STM32F407ZETx', 'Untitled - Project Manager', and a 'GENERATE CODE' button. Below this, there are four tabs: 'Pinout & Configuration', 'Clock Configuration', 'Project Manager' (which is active), and 'Tools'. On the left side of the 'Project Manager' tab, there are two sub-tabs: 'Project' and 'Code Generator'. The 'Project' sub-tab is selected, displaying 'Project Settings'. The settings include: 'Project Name' (sram_cubemx), 'Project Location' (D:\project\2026\Mango-M32F4 교육\PPT\9.SRAM\), 'Application Structure' (Advanced), 'Toolchain Folder Location' (D:\project\2026\Mango-M32F4 교육\PPT\9.SRAM\), and 'Toolchain / IDE' (STM32CubeIDE). A checkbox for 'Generate Under Root' is checked.

Project Settings	Value
Project Name	sram_cubemx
Project Location	D:\project\2026\Mango-M32F4 교육\PPT\9.SRAM\
Application Structure	Advanced
Toolchain Folder Location	D:\project\2026\Mango-M32F4 교육\PPT\9.SRAM\
Toolchain / IDE	STM32CubeIDE

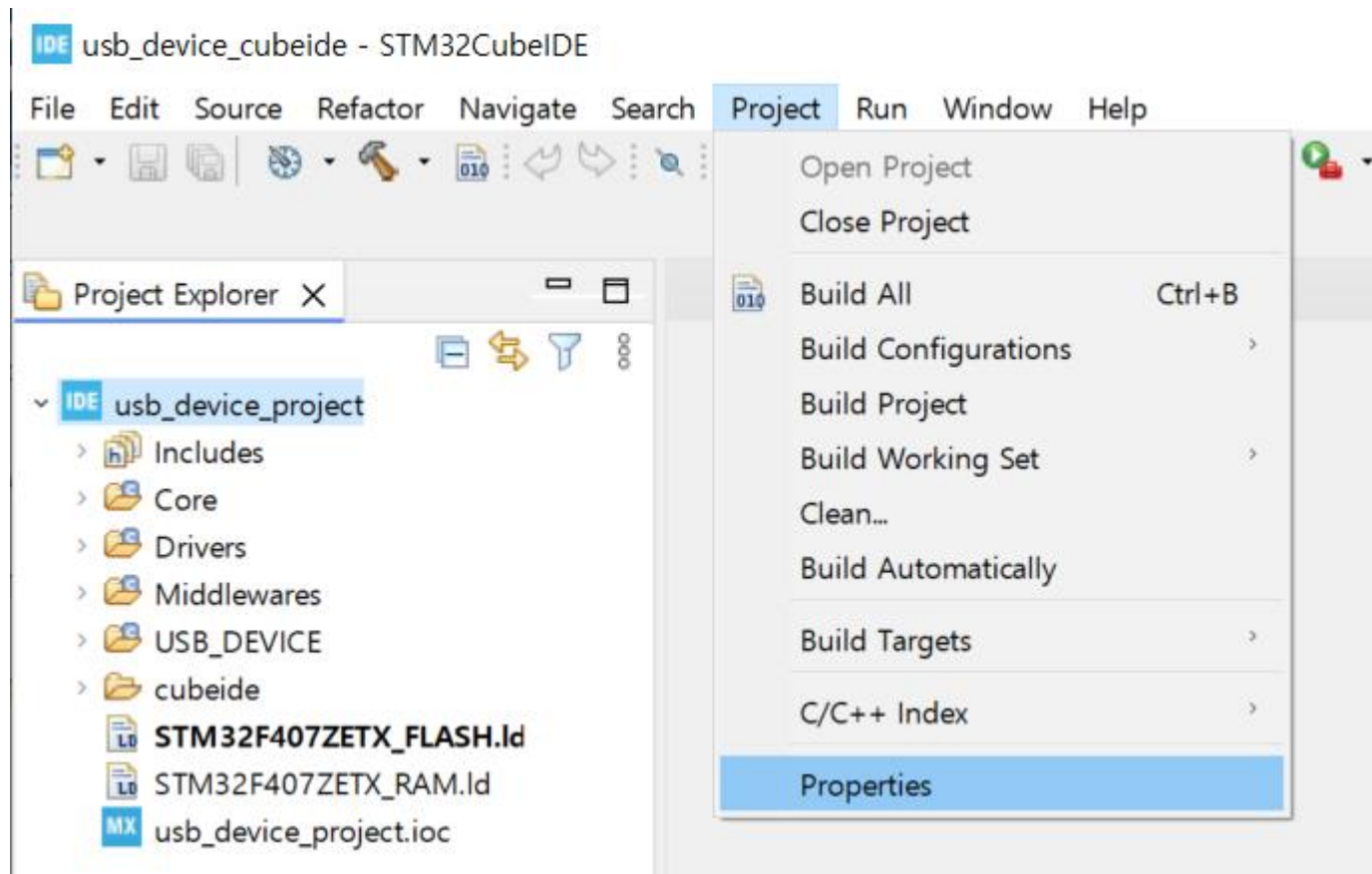
Project 생성

- Open Project 클릭하여 STM32CubeIDE 실행, workspace는 CubeMX와 다른 폴더로 지정



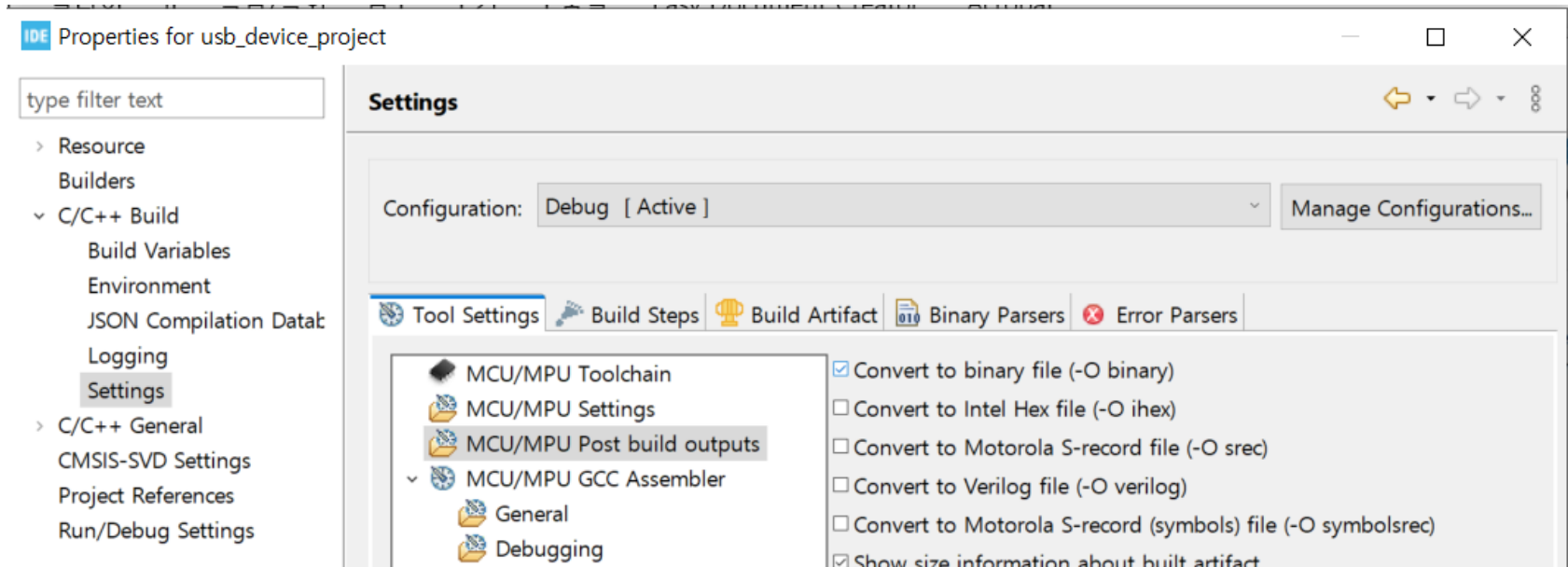
BIN 파일 생성 설정

- STM32CubeProgrammer로 시리얼 다운로드하기 위해서는 BIN파일이 필요.
- Project -> Properties 클릭



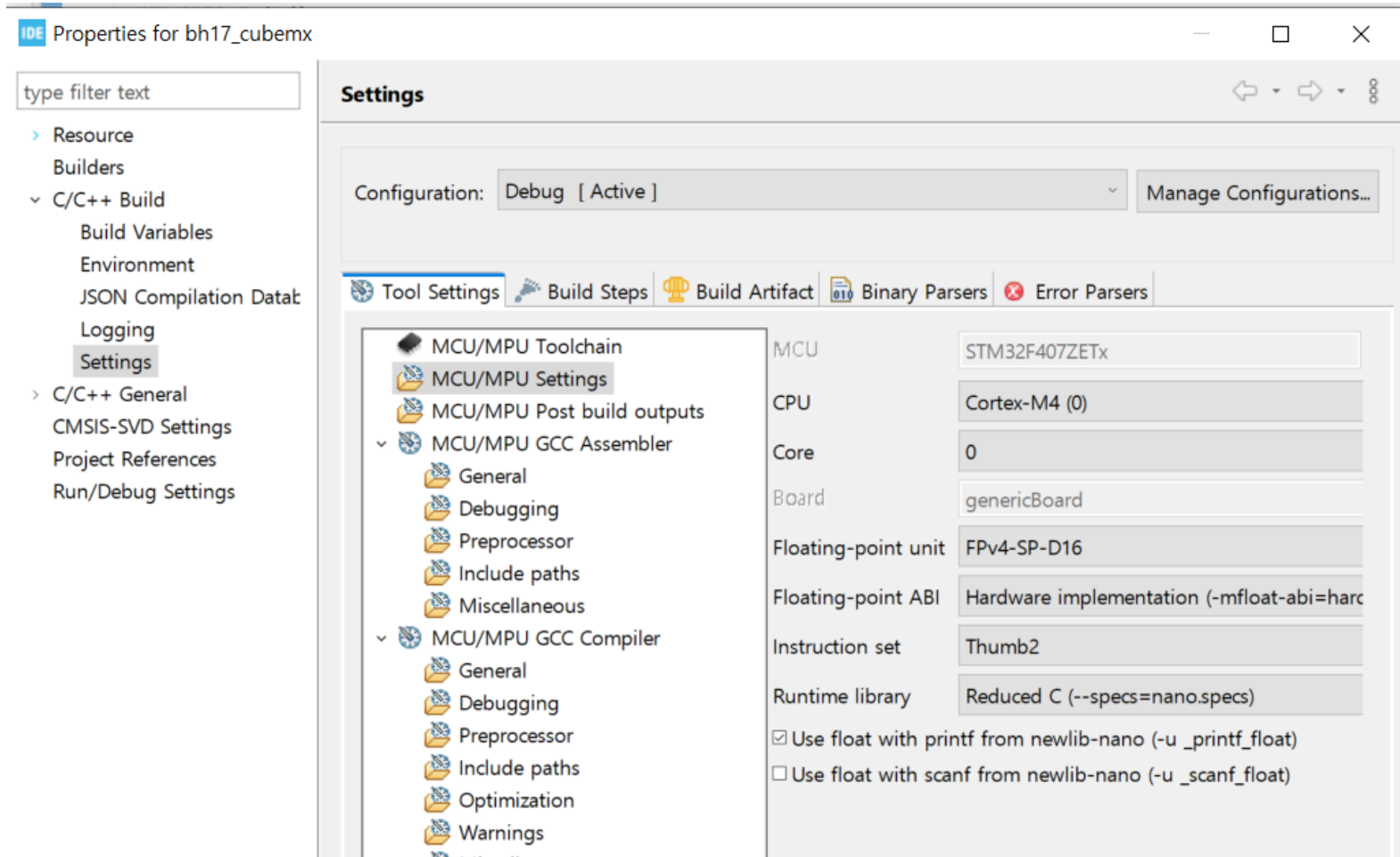
BIN 파일 생성 설정

- C/C++ Build -> Settings -> Tool Settings -> MCU/MPU Post build outputs => Convert to binary file(-O binary) 클릭.



빌드 설정

- C/C++ Build -> Settings -> Tool Settings -> MCU/MPU Settings => Use float with printf from newlib-nano(-u _printf_float) 클릭.



BNO055 데이터시트

- Slave Address : 0x28

Table 4-7: I2C address selection

I2C configuration	COM3_state	I2C address
Slave	HIGH	0x29
Slave	LOW	0x28
HID-I2C	X	0x40

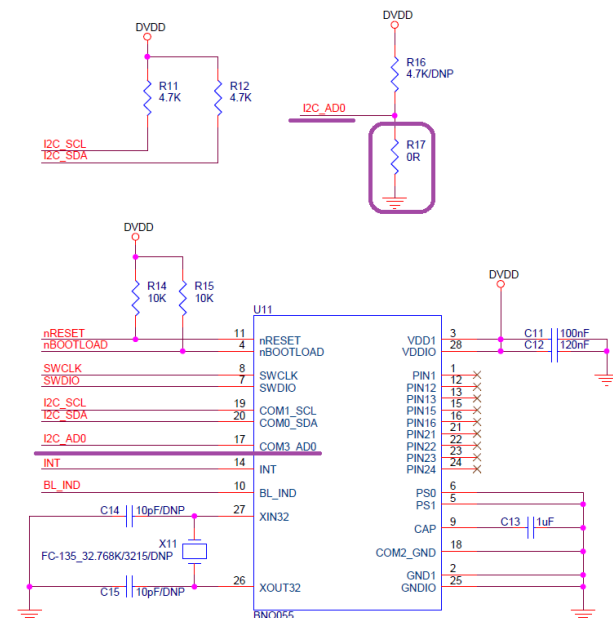
- CHIP_ID_REG : 0x00

4.3.1 CHIP_ID 0x00

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r	r	r	r	r	r	r	r
Reset	1	0	1	0	0	0	0	0
Content	BNO055 CHIP ID							

DATA	bits	Description
BNO055 CHIP ID	<7:0>	Chip identification code, read-only fixed value 0xA0

9 AXIS_SENSOR



- S/W reset : 0x3F, 0x20

4.3.63 SYS_TRIGGER 0x3F

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	w	r/w	r/w	r/w	r/w	w
Reset	0	0	0	0	0	0	0	0
Content	CLK_SEL	RST_INT	RST_SYS	Reserved				Self_Test

DATA	bits	Description
CLK_SEL	7	0: Use internal oscillator 1: Use external oscillator. Set this bit only if external crystal is connected
RST_INT	6	Set to reset all interrupt status bits, and INT output
RST_SYS	5	Set to reset system
Self_Test	0	Set to trigger self-test

- OPR Mode : 0x3D

4.3.61 OPR_MODE 0x3D

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	0	0	0	1	0	0	0	0
Content	Reserved				Operation Mode <3:0>			

DATA	bits	Description
Operation Mode <3:0>	<3:0>	Read: Current selected operation mode Write: Select operation mode See Table 3-5 for details

Table 3-5: operating modes selection

Parameter	Value	[Reg Addr]: Reg Value
CONFIG MODE	CONFIGMODE	[OPR_MODE]: xxxx0000b
Non-Fusion Mode	ACCONLY	[OPR_MODE]: xxxx0001b
	MAGONLY	[OPR_MODE]: xxxx0010b
	GYROONLY	[OPR_MODE]: xxxx0011b
	ACCMAG	[OPR_MODE]: xxxx0100b
	ACCGYRO	[OPR_MODE]: xxxx0101b
	MAGGYRO	[OPR_MODE]: xxxx0110b
Fusion Mode	AMG	[OPR_MODE]: xxxx0111b
	IMU	[OPR_MODE]: xxxx1000b
	COMPASS	[OPR_MODE]: xxxx1001b
	M4G	[OPR_MODE]: xxxx1010b
	NDOF_FMC_OFF	[OPR_MODE]: xxxx1011b
	NDOF	[OPR_MODE]: xxxx1100b

- Power Mode : 0x3E

4.3.62 PWR_MODE 0x3E

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	0	0	0	0	0	0	0	0
Content	Reserved					Power Mode <1:0>		

DATA	bits	Description
Power Mode <1:0>	<1:0>	Read: Current selected power mode Write: Select power mode See Table 3-1 for details

Table 3-1: power modes selection

Parameter	Value	[Reg Addr]: Reg Value
Power Mode	Normal Mode	[PWR_MODE]: xxxxxx00b
	Low Power Mode	[PWR_MODE]: xxxxxx01b
	Suspend Mode	[PWR_MODE]: xxxxxx10b
	Invalid	[PWR_MODE]: xxxxxx11b

- NDOF Mode : 0x3D

4.3.61 OPR_MODE 0x3D

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	0	0	0	1	0	0	0	0
Content	Reserved				Operation Mode <3:0>			

DATA	bits	Description
Operation Mode <3:0>	<3:0>	Read: Current selected operation mode Write: Select operation mode See Table 3-5 for details

Table 3-3: Operating modes overview

Operating Mode		Available sensor signals			Fusion Data	
		Accel	Mag	Gyro	Relative orientation	Absolute orientation
Non-fusion modes	CONFIGMODE	-	-	-	-	-
	ACCONLY	X	-	-	-	-
	MAGONLY	-	X	-	-	-
	GYROONLY	-	-	X	-	-
	ACCMAG	X	X	-	-	-
	ACCGYRO	X	-	X	-	-
	MAGGYRO	-	X	X	-	-
Fusion modes	AMG	X	X	X	-	-
	IMU	X	-	X	X	-
	COMPASS	X	X	-	-	X
	M4G	X	X	-	X	-
	NDOF_FMC_OFF	X	X	X	-	X
	NDOF	X	X	X	-	X

- Power Mode : 0x3E

4.3.62 PWR_MODE 0x3E

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	0	0	0	0	0	0	0	0
Content	Reserved					Power Mode <1:0>		

DATA	bits	Description
Power Mode <1:0>	<1:0>	Read: Current selected power mode Write: Select power mode See Table 3-1 for details

Table 3-1: power modes selection

Parameter	Value	[Reg Addr]: Reg Value
Power Mode	Normal Mode	[PWR_MODE]: xxxxxx00b
	Low Power Mode	[PWR_MODE]: xxxxxx01b
	Suspend Mode	[PWR_MODE]: xxxxxx10b
	Invalid	[PWR_MODE]: xxxxxx11b

- NDOF Mode : 0x3D

4.3.61 OPR_MODE 0x3D

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	0	0	0	1	0	0	0	0
Content	Reserved				Operation Mode <3:0>			

DATA	bits	Description
Operation Mode <3:0>	<3:0>	Read: Current selected operation mode Write: Select operation mode See Table 3-5 for details

Table 3-3: Operating modes overview

Operating Mode		Available sensor signals			Fusion Data	
		Accel	Mag	Gyro	Relative orientation	Absolute orientation
Non-fusion modes	CONFIGMODE	-	-	-	-	-
	ACCONLY	X	-	-	-	-
	MAGONLY	-	X	-	-	-
	GYROONLY	-	-	X	-	-
	ACCMAG	X	X	-	-	-
	ACCGYRO	X	-	X	-	-
	MAGGYRO	-	X	X	-	-
Fusion modes	AMG	X	X	X	-	-
	IMU	X	-	X	X	-
	COMPASS	X	X	-	-	X
	M4G	X	X	-	X	-
	NDOF_FMC_OFF	X	X	X	-	X
	NDOF	X	X	X	-	X

- UNIT_SEL : 0x3B

4.3.60 UNIT_SEL 0x3B

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Reset	1	0	0	0	0	0	0	0
Content	ORI_Android_Window s	reserved		TEMP_Unit	reserved	EUL_Unit	GYR_Unit	ACC_Unit

Table 3-11: unit selection

Data	Units	[Reg Addr]: Register Value
Acceleration	m/s ²	[UNIT_SEL] : xxxxxxx0b
	mg	[UNIT_SEL] : xxxxxxx1b
Linear Acceleration, Gravity vector	m/s ²	[UNIT_SEL] : xxxxxxx0b
Magnetic Field Strength	Micro Tesla	NA
Angular Rate	Dps	[UNIT_SEL] : xxxxxx0xb
	Rps	[UNIT_SEL] : xxxxxx1xb
Euler Angles	Degrees	[UNIT_SEL] : xxxxx0xxb
	Radians	[UNIT_SEL] : xxxxx1xxb
Quaternion	Quaternion units	NA
Temperature	°C	[UNIT_SEL] : xxx0xxxxb
	°F	[UNIT_SEL] : xxx1xxxxb

Table 3-29: Euler angle data representation

Unit	Representation
Degrees	1 degree = 16 LSB
Radians	1 radian = 900 LSB

Table 3-33: Linear Acceleration data representation

Unit	Representation
m/s^2	1 m/s^2 = 100 LSB
mg	1 mg = 1 LSB

Table 3-37: Temperature data representation

Unit	Representation
$^{\circ}\text{C}$	1 $^{\circ}\text{C}$ = 1 LSB
F	2 F = 1 LSB

- Euler angels (0x1A ~ 0x1F)

1F	EUL_Pitch_MSB	0x00	Pitch Data <15:8>
1E	EUL_Pitch_LSB	0x00	Pitch Data <7:0>
1D	EUL_Roll_MSB	0x00	Roll Data <15:8>
1C	EUL_Roll_LSB	0x00	Roll Data <7:0>
1B	EUL_Heading_MSB	0x00	Heading Data <15:8>
1A	EUL_Heading_LSB	0x00	Heading Data <7:0>

- Quaternion (0x20 ~ 0x27)

27	QUA_Data_z_MSB	0x00	Quaternion z Data <15:8>
26	QUA_Data_z_LSB	0x00	Quaternion z Data <7:0>
25	QUA_Data_y_MSB	0x00	Quaternion y Data <15:8>
24	QUA_Data_y_LSB	0x00	Quaternion y Data <7:0>
23	QUA_Data_x_MSB	0x00	Quaternion x Data <15:8>
22	QUA_Data_x_LSB	0x00	Quaternion x Data <7:0>
21	QUA_Data_w_MSB	0x00	Quaternion w Data <15:8>
20	QUA_Data_w_LSB	0x00	Quaternion w Data <7:0>

- Accelerometer (0x8 ~ 0xD) : w/ Gravity

D	ACC_DATA_Z_MSB	0x00	Acceleration Data Z <15:8>
C	ACC_DATA_Z_LSB	0x00	Acceleration Data Z <7:0>
B	ACC_DATA_Y_MSB	0x00	Acceleration Data Y <15:8>
A	ACC_DATA_Y_LSB	0x00	Acceleration Data Y <7:0>
9	ACC_DATA_X_MSB	0x00	Acceleration Data X <15:8>
8	ACC_DATA_X_LSB	0x00	Acceleration Data X <7:0>

- Gyroscope (0x14 ~ 0x19)

19	GYR_DATA_Z_MSB	0x00	Gyroscope Data Z <15:8>
18	GYR_DATA_Z_LSB	0x00	Gyroscope Data Z <7:0>
17	GYR_DATA_Y_MSB	0x00	Gyroscope Data Y <15:8>
16	GYR_DATA_Y_LSB	0x00	Gyroscope Data Y <7:0>
15	GYR_DATA_X_MSB	0x00	Gyroscope Data X <15:8>
14	GYR_DATA_X_LSB	0x00	Gyroscope Data X <7:0>

- Accelerometer (0x28 ~ 0x2D) : w/o Gravity

2D	LIA_Data_Z_M BS	0x00	Linear Acceleration Data Z <15:8>
2C	LIA_Data_Z_LS B	0x00	Linear Acceleration Data Z <7:0>
2B	LIA_Data_Y_M BS	0x00	Linear Acceleration Data Y <15:8>
2A	LIA_Data_Y_LS B	0x00	Linear Acceleration Data Y <7:0>
29	LIA_Data_X_M BS	0x00	Linear Acceleration Data X <15:8>
28	LIA_Data_X_LS B	0x00	Linear Acceleration Data X <7:0>

- Temperature : 0x34

34	TEMP	0x00	Temperature
----	------	------	-------------

Code 수정

- UART1으로 로그를 내 보내기 위해 아래 코드 추가 :
core/src/main_example.c

```
/* printf → UART1 리디렉션 */  
int _write(int file, char *ptr, int len)  
{  
    HAL_UART_Transmit(&huart1, (uint8_t *)ptr, (uint16_t)len, 100);  
    return len;  
}
```

- BNO055의 slave address는 0x28이다.

```
/* _____  
 * I2C 주소  
 * COM3 핀 High → 0x29 (default)  
 * COM3 핀 Low → 0x28  
 * _____*/  
#define BNO055_I2C_ADDR_HIGH (0x29 << 1) /* COM3 = HIGH (default) */  
#define BNO055_I2C_ADDR_LOW (0x28 << 1) /* COM3 = LOW  
*/
```

- 초기화시에 Software Reset를 한다.

```
BNO055_Status_t BNO055_Init(BNO055_Handle_t *hbno,
                             I2C_HandleTypeDef *hi2c,
                             uint8_t i2c_addr,
                             bool use_ext_crystal)
{
    BNO055_Status_t ret;
    uint8_t chip_id;

    /* — CHIP_ID 확인 (0xA0 고정) — */
    ret = bno_read_reg(hbno, BNO055_CHIP_ID_REG, &chip_id);

    /* — 소프트웨어 리셋 — */
    ret = BNO055_Reset(hbno);

    /* — CONFIG 모드 진입 확인 — */
    ret = bno_set_mode_internal(hbno, BNO055_MODE_CONFIG);

    /* — 전원 모드: Normal — */
    ret = bno_write_reg(hbno, BNO055_PWR_MODE_REG, BNO055_POWER_NORMAL);

    /* — 단위 설정: m/s², dps, degree, °C, Windows 좌표계 —
       UNIT_SEL[7]=0: Windows [4]=0: °C
       [2]=0: degree [1]=0: dps [0]=0: m/s² */
    ret = bno_write_reg(hbno, BNO055_UNIT_SEL_REG, 0x00);

    /* — NDOF 모드 진입 (9축 Sensor Fusion) — */
    ret = bno_set_mode_internal(hbno, BNO055_MODE_NDOF);
    if (ret != BNO055_OK) return ret;

    return BNO055_OK;
}
```

Code 수정

- 1초 주기로 센서 데이터를 표시한다.

```
While (1) {
    /* 오일러각 (Fusion 출력) */
    if (BNO055_GetEuler(&bno, &euler) == BNO055_OK) {
        printf("Euler | Heading=%6.2f Roll=%6.2f Pitch=%6.2f [deg]\r\n",
            (double)euler.heading, (double)euler.roll, (double)euler.pitch);
    }

    /* 쿼터니언 (Fusion 출력) */
    if (BNO055_GetQuaternion(&bno, &quat) == BNO055_OK) {
        printf("Quat | W=%6.4f X=%6.4f Y=%6.4f Z=%6.4f\r\n", (double)quat.w, (double)quat.x, (double)quat.y,
            (double)quat.z);
    }

    /* 가속도 (중력 포함) */
    if (BNO055_GetAccel(&bno, &accel) == BNO055_OK) {
        printf("Accel | X=%6.2f Y=%6.2f Z=%6.2f [m/s^2]\r\n", (double)accel.x, (double)accel.y, (double)accel.z);
    }

    /* 자이로 */
    if (BNO055_GetGyro(&bno, &gyro) == BNO055_OK) {
        printf("Gyro | X=%7.2f Y=%7.2f Z=%7.2f [dps]\r\n", (double)gyro.x, (double)gyro.y, (double)gyro.z);
    }

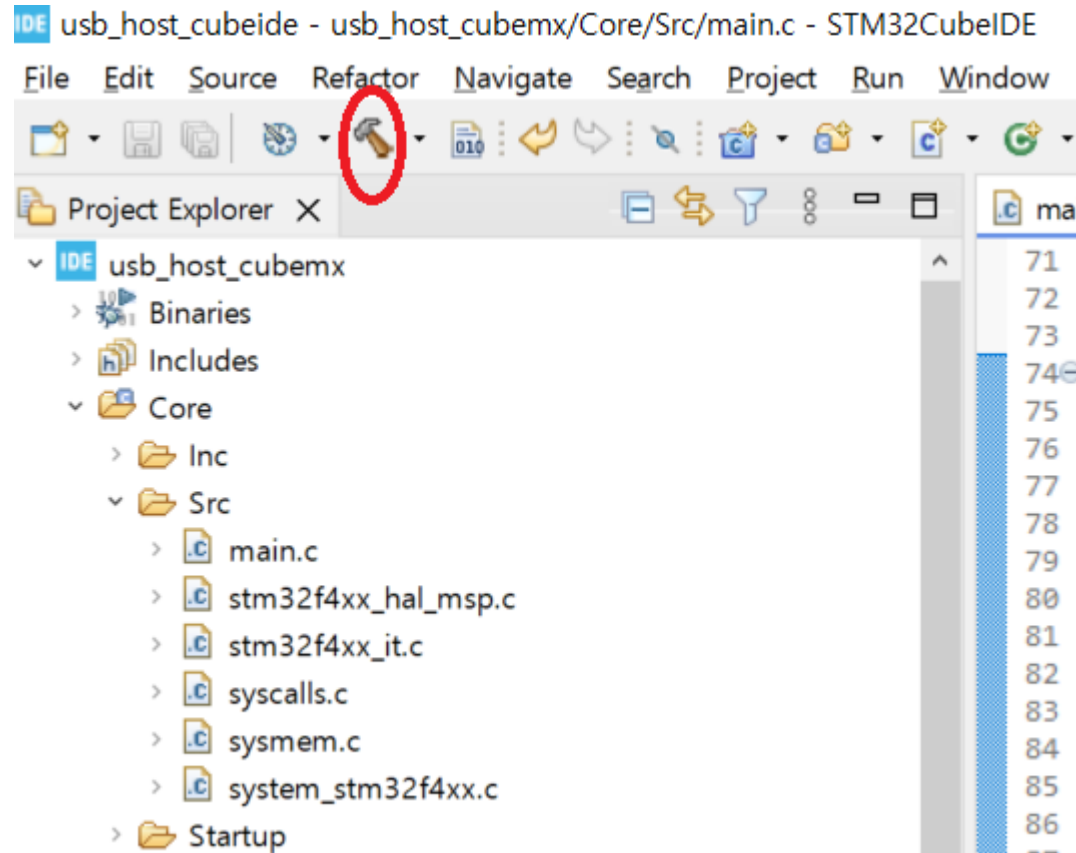
    /* 선형가속도 (중력 제거, Fusion 출력) */
    if (BNO055_GetLinearAccel(&bno, &lia) == BNO055_OK) {
        printf("LinAcc | X=%6.2f Y=%6.2f Z=%6.2f [m/s^2]\r\n", (double)lia.x, (double)lia.y, (double)lia.z);
    }

    /* 온도 */
    if (BNO055_GetTemperature(&bno, &temp) == BNO055_OK) {
        printf("Temp | %d [°C]\r\n", temp);
    }

    HAL_Delay(1000); /* 1 Hz 출력 */
}
```

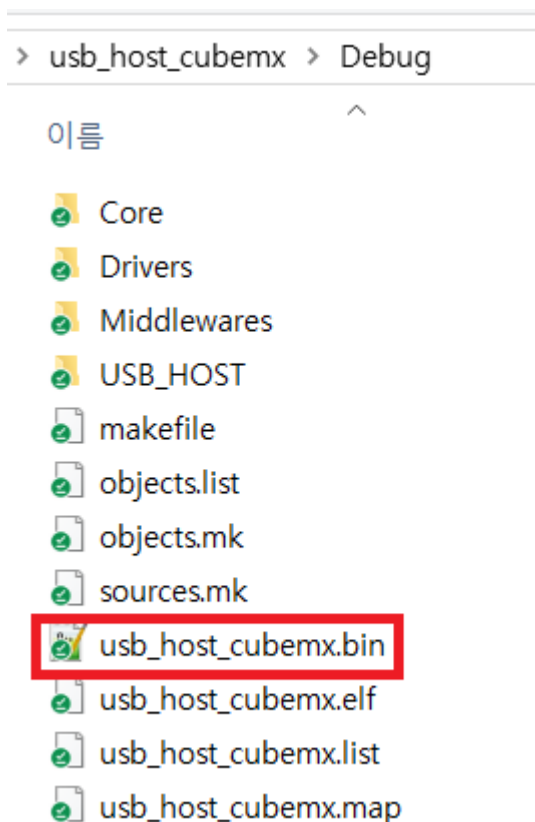
Build

- 아래 버튼 클릭하여 Build



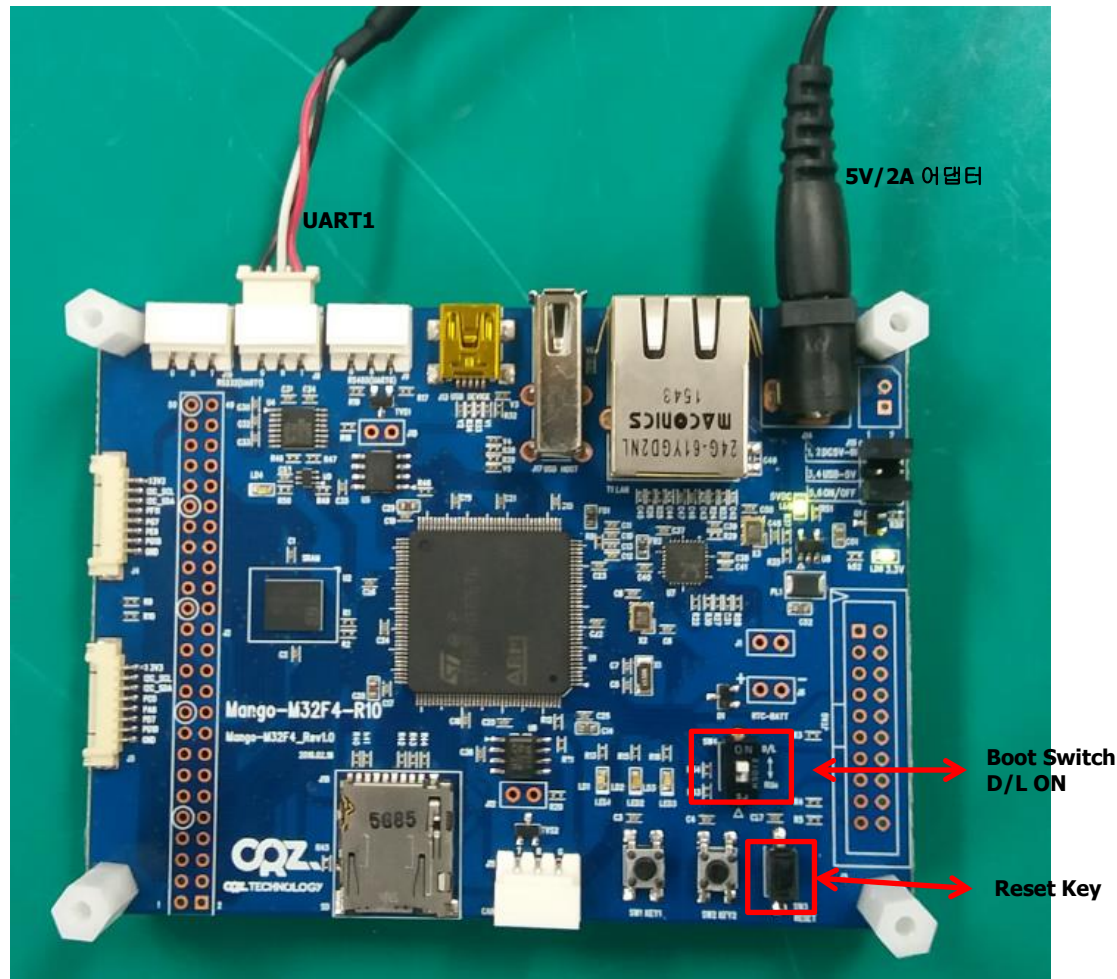
Build 결과물

- 빌드 결과물 이름은 [프로젝트 이름].bin이다.



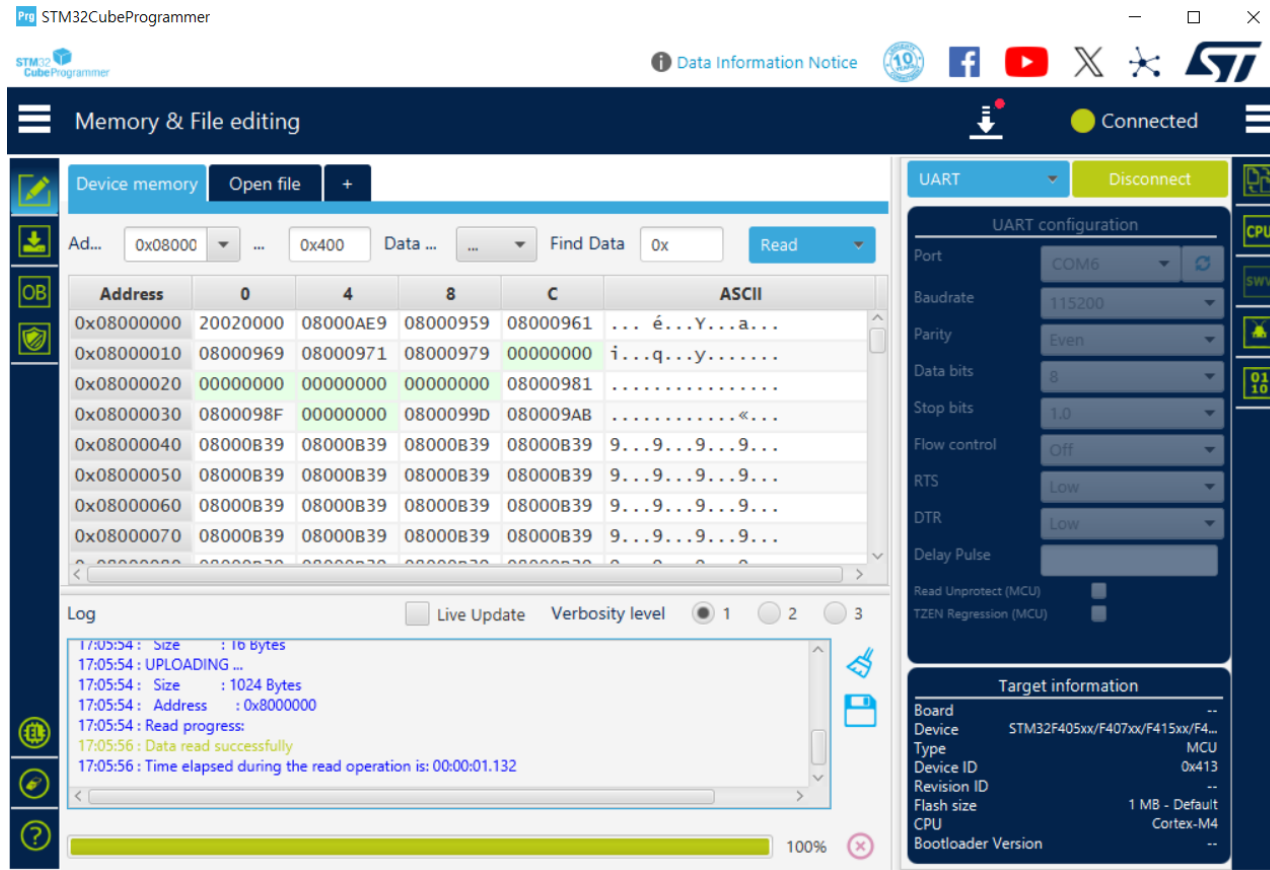
BIN 다운로드

- UART1과 연결되어 있는 터미널(Teraterm) 종료.
- Mango-M32F4를 다운로드 모드로 설정하고 SW3 리셋 키를 누른다.



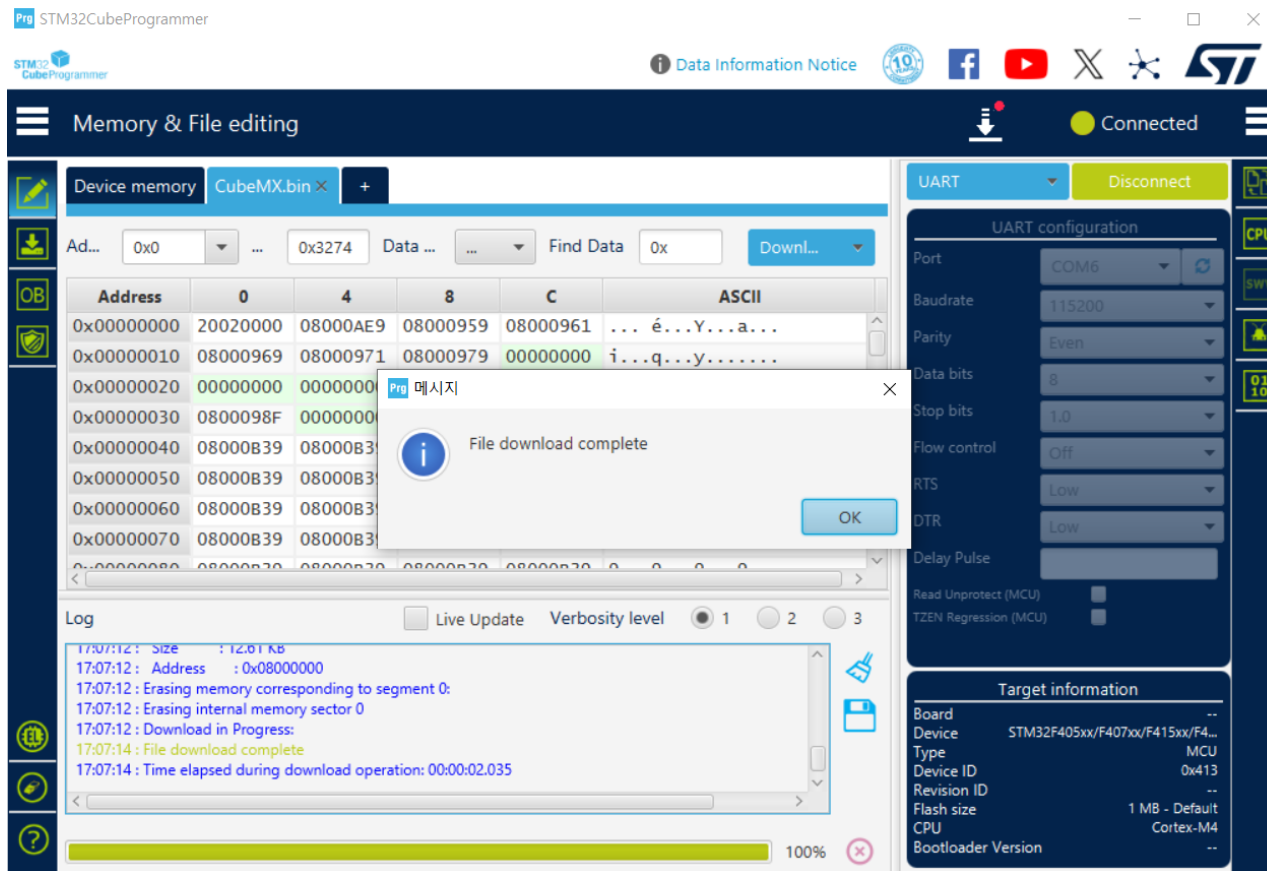
BIN 다운로드

- STM32CubeProgrammer를 실행한다.



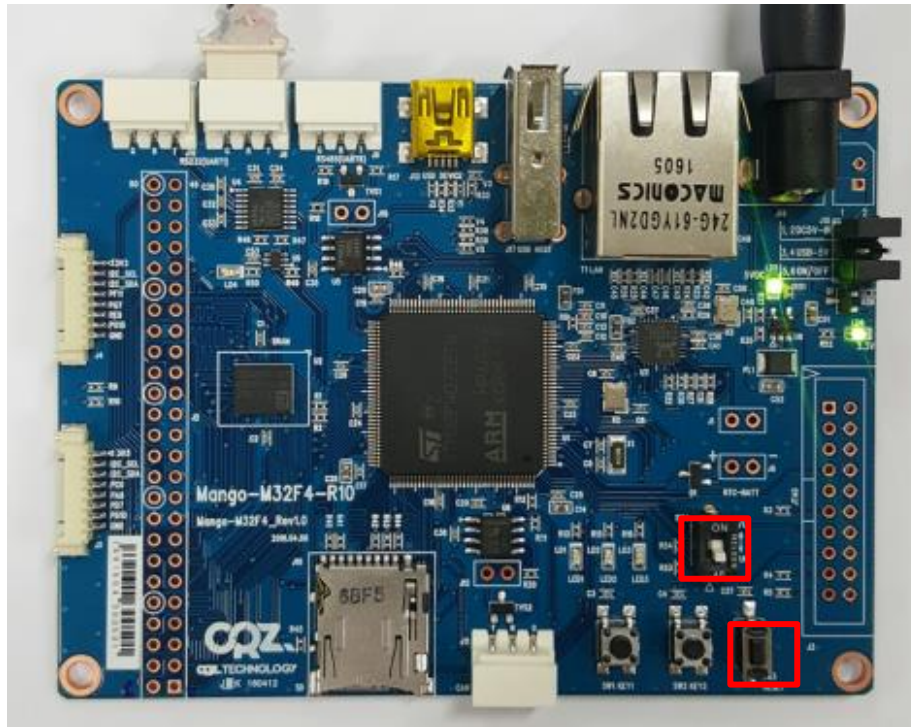
BIN 다운로드

- 빌드된 BIN 파일을 열어서 다운로드한다.



실행결과

- STM32CubeProgrammer의 COM Port를 연결해제하고 터미널 (Teraterm)을 실행한다.
- Mango-M32F4를 실행모드로 설정하고 SW3 리셋 키를 누른다.



실행결과

- CS-BNO055 센서 보드를 Mango-M32F4 보드의 J4 혹은 J5에 8핀 커넥터를 이용하여 연결한다.
- 9축 센서 데이터가 정상적으로 읽힘.

```
COM6 - Tera Term VT
File Edit Setup Control Window Help

===== BNO055 STM32F4 I2C Example =====
[OK] BNO055 initialized (NDOF mode)
SYS_STATUS=0x05 SYS_ERR=0x00 ST_RESULT=0x0F

--- Sensor Data ---
Euler | Heading= 0.00 Roll= 0.00 Pitch= 0.00 [deg]
Quat  | W=0.0000 X=0.0000 Y=0.0000 Z=0.0000
Accel | X= -1.62 Y= 9.29 Z= 2.62 [m/s²]
Gyro  | X= -0.25 Y= -0.12 Z= 0.31 [dps]
LinAcc| X= 0.00 Y= 0.00 Z= 0.00 [m/s²]
Temp  | 30 [°C]

Euler | Heading=359.94 Roll= -9.56 Pitch= 74.12 [deg]
Quat  | W=0.7968 X=0.5952 Y=0.1048 Z=0.0001
Accel | X= -1.56 Y= 8.87 Z= 2.48 [m/s²]
Gyro  | X= 0.19 Y= -0.19 Z= 0.00 [dps]
LinAcc| X= 0.08 Y= -0.43 Z= -0.15 [m/s²]
Temp  | 30 [°C]

Euler | Heading=359.94 Roll= -9.56 Pitch= 74.00 [deg]
Quat  | W=0.7973 X=0.5944 Y=0.1047 Z=0.0002
Accel | X= -1.62 Y= 9.26 Z= 2.61 [m/s²]
Gyro  | X= -0.06 Y= -0.31 Z= 0.19 [dps]
LinAcc| X= 0.02 Y= -0.03 Z= -0.03 [m/s²]
Temp  | 30 [°C]
```